

Frederico Ramos

Curriculum Vitae

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DBLP

PhD candidate in program analysis and automated reasoning: specification-driven symbolic execution, constraint solving (Z3 and other SMT/SAT solvers), and summary generation for program verification.

Education

- 2022–present **PhD in Computer Science and Engineering**, *Instituto Superior Técnico*, Lisbon, Portugal
2019–2021 **M.Sc. in Information Systems and Computer Engineering**, *Instituto Superior Técnico*
Thesis Toward Tool-Independent Summaries for Symbolic Execution (Grade: 19/20)
Supervisors Prof. Pedro Adão and Prof. José Frago
2015–2019 **B.Sc. in Information Systems and Computer Engineering**, *Instituto Superior Técnico*

Professional Experience

- Summer 2024 **Software Engineering Intern**, Coana Tech, Aarhus, Denmark
Extended an existing dependency-graph-based SCA tool with support for multiple package managers (npm, pnpm, yarn, Go modules) to detect and fix vulnerable packages.
2022–2026 **Teaching Assistant**, Instituto Superior Técnico, DEI
Courses: Forensics Cyber-Security; Operating Systems; Distributed Systems; Analysis and Synthesis of Algorithms; and Software Engineering

Publications

- 2026 **ESOP'26**, CORE A, *Specification-Driven Generation of Summaries for Symbolic Execution*
2023 **ECOOP'23**, CORE A, *Toward Tool-Independent Summaries for Symbolic Execution*
2023 **ISSTA'23**, CORE A, *MetaData262: Automatic Test Suite Selection for Partial JavaScript Implementations*
2022 **KLEE'22**, Poster at the 3rd International KLEE Workshop on Symbolic Execution
Empirical Study on Applying Program Analysis and Testing Tools to Student Code

Invited Talks

- 2023 **Amazon**, Invited Talk at Amazon's London office hosted by Daniel Schoepe
Automatic Test Suite Selection for Partial JavaScript Implementations
2022 **Meta**, Invited Talk at Meta's London office hosted by Jules Villard
Toward Tool-Independent Summaries for Symbolic Execution

Extra Activities

- 2023 **MOD23**, Summer School Marktoberdorf 2023
2023 **Student Volunteer**, ECOOP'23 & ISSTA'23 Seattle
2022 **Student Volunteer**, ECOOP'22 Berlin

Awards & Honors

- 2023 **Excellence Award for Outstanding Teachers**, Instituto Superior Técnico
(Academic year 2022/2023)
2022 **Academic Merit Diploma**, Instituto Superior Técnico
(Academic year 2020/2021)

Technical Skills

- Programming Python, TypeScript, OCaml, C, SMT-LIB
Reasoning Constraint solving & SMT (extensive experience with Z3 and other SMT/SAT solvers), symbolic execution, program analysis, test generation; and exposure to Dafny